

Episode B3 | *Equilibrium*

Equilibrium: when no one wants to change

Last week we started with buyers: we found that individuals preferences about how much spinach to buy obey the law of demand giving us an individual demand curve. Then we took many buyers and found the spinach demand curve. Representing this with algebra allowed us to find the quantity demanded at a whole range of prices.

Then we turned to the other side of the market, sellers: we allowed Molly to make her decision about where to live on the frontier which gave us her individual supply curve, which obeys the law of supply. Then we took all the many farmers growing spinach and found the spinach supply curve. Representing this with algebra allowed us to find the quantity supplied at a whole range of prices.

This answered part of the question about where on the frontier to live but introduced another free variable left to float around Price. In some sense we simply pushed the question of where to live on the PPF one level deeper into prices. Now we put the two together and ask whether there's a price at which those quantities agree. So this brings us to the question. What happens to prices when we have supply and demand in a market for spinach? Let's run a little fun simulation to build up some intuition for markets.

To do this, let's set up a farmer's market for spinach and use two graphs. One for supply and one for demand.

{++Let's start with just two people. Gary would pay up to \$6 for a pound of spinach. Growing that pound costs Molly \$2. Now suppose a price of \$4 comes up between them. Would they both say yes? Gary pays less than the spinach is worth to him. Molly receives more than it cost her. So the trade happens. And notice that any price between \$2 and \$6 would have worked — the same window we saw in Part A, when our two farmers settled on an exchange rate. ++}

{++Let's zoom way in on this one exchange. Gary hands over \$4 — his expenditure. Against his marginal benefit of \$6, that leaves him \$2 of consumer surplus. Molly receives the same \$4 as revenue. Above her cost of \$2, that leaves her \$2 of producer surplus. The price is doing the splitting: it divides the gains from this trade between the two of them.++}

{++Now let's add another buyer. Two buyers want spinach and Molly only has so much. Whoever would go home empty-handed has a reason to offer a little more, so the price creeps up. Add another seller and the pressure runs the other way: sellers competing for buyers cut their prices. We can keep adding people this way, one at a time, and a market takes shape.++}

{++With ten buyers and ten sellers we can let the market run. Buyers check prices and switch to better deals. Sellers stuck with spinach lower their asks. Buyers stuck in line offer more.++}

{++Watch what happens: the trading gets calmer and calmer, until the prices stop moving. Nobody told anyone to stop. No one wants to change.++}

Can there ever be a price that doesn't change? Yes. What if quantity demanded was equal to quantity supplied? Here no one has an incentive to change their prices or quantities. If a seller raised their price, they would lose all their business.

If they lower their price, they would make less than they could. If a buyer raised their price, they would be paying more than they needed to. And if the buyer lowered their price no seller would sell to them. In this way incentives push the price to the point where quantity supplied is equal to quantity demanded.

{++***Equilibrium:*** *The price and quantity at which quantity supplied equals quantity demanded — where no one wants to change.*++}

Here we're going to use algebra to build a model using both supply and demand to arrive at the prices that coordinate sellers choices of quantity supplied and buyers choices of quantity demanded.

The supply curve for spinach was price is equal to two plus one over twenty quantity and the demand curve for spinach was price is equal to twelve minus one over five quantity. We can plot them on the same graph. There's one price that allows quantity supplied to equal quantity demanded. And that's where they intersect.

Let's first solve for the equilibrium quantity then solve for equilibrium price. We set the supply curve equal to the demand curve. Then we solve for quantity which gives us 40. We then take this and plug it into the supply curve.

And solve for price which comes out to be \$4. It would be equivalent if you plugged the quantity into the demand curve and solved for price that way but plugging it into the supply curve is usually easier.

We denote the equilibrium price and quantity with a star.

We typically combine supply and demand on one graph, so let's do that here.

And to fully characterize equilibrium on this graph, we will solve for P^* and Q^* . Plot it on the graph.

{++Pause for Exercise B3 / Q1: solve for Q^ and P^* in the pumpkin-pasty market on the exercise sheet, then plot the starred pair.++}*

{++But suppose the buyers aren't happy paying \$4. A bunch of them get together and insist on a lower price: \$3 a pound.++}

Let's say spinach sellers of which Molly and Andrew are included decide to sell spinach at a price of \$3. We can find the quantity supplied at this price. Which comes out to be 20,000 pounds. But is this how much buyers are willing to buy at this price? The quantity demanded is 45,000 pounds and we find this by plugging in price into the demand curve.

So it turns out people want more than is available which we call a shortage. But let's say in the situation. Amanda-Grace an entrepreneurial buyer decides to offer Molly a higher price. Let's say three dollars and twenty-five cents. This is

good for Molly since she's making more and it's good for Amanda-Grace who is more than willing to pay for the higher price if she can receive what she wants.

{++**Shortage:** *Quantity demanded is greater than quantity supplied.*++}

But it turns out Gary who is buying from Molly now can't get any spinach. So, what should he do? He should raise the price.

In this way, whenever there's a shortage both buyers and sellers have an incentive to raise their prices.

{++Of course, the sellers can play the same game in the other direction.++}

Let's say instead spinach sellers sell at a price of \$6. At this price the quantity supplied is 80,000 pounds. And the quantity demanded is 30,000. So people want less than is available. Which we'll call an excess.

{++**Excess:** *Quantity supplied is greater than quantity demanded.*++}

I want to offer a little side note here most textbooks call this a surplus but for reasons we'll get into later. I think this naming convention is less than ideal and can be confusing. So instead of surplus we use excess to describe quantity supplied being greater than quantity demanded. But I just want you to be aware that it's often called surplus.

So Molly and Andrew are sitting around with excess spinach. They can't sell at 6 per pound. So Molly being the entrepreneur farmer, decides to offer a price of \$5 to Gary.

Gary was buying spinach from Andrew but is more than willing to buy spinach from Molly at the lower price. And would even buy a little extra. You can think of the law of demand here. Andrew isn't happy about losing Gary's business. So he decides to lower his prices too.

In this way, whenever there's excess both buyers and sellers have an incentive to lower their prices.

~~Okay. ~~So prices will rise with a shortage and lower with an excess.

Okay, so we've found where prices come from. And we've found an equilibrium concept that's stable.

At this equilibrium price and quantity no buyer and no seller wants to change the price they're offering and the quantity they're either supplying or demanding from the market. So we call this equilibrium stable.

{++Notice what happened both times: a whole side of the market teamed up to move the price, and it still didn't hold. Someone always found it worthwhile to break ranks.++}

{++Pause for Exercise B3 / Q2: at 5 galleons on the pumpkin-pasty curves, find the quantity demanded and the quantity supplied, name the situation, and predict which way the price moves.++}

PPF Tieback

We've spent time talking about markets, the preferences of buyers and sellers, and the incentives that push markets toward equilibrium. And we found that this equilibrium gave us two things: a price and a quantity.

At the beginning of the semester we began by framing the class in the context of the PPF. We had our two farmers who found that if they specialized and traded with each other, they could do better than they could on their own (in autarky).

But we were left with the question of where on the PPF to live. If we fully specialized, we knew what the point on the PPF would be. But we wouldn't necessarily always fully specialize. We could partially specialize, and still be outside the autarky PPF. So that was a question.

And we proposed markets as a way of picking the point, the pair of quantities. So we built up the intuition for the demand curve, which tells us how much

buyers will demand at all possible prices, which satisfies the law of demand. And we built the supply curve, which satisfies the law of supply.

Then we developed our intuition for the market equilibrium. Buyers and sellers have incentive to move the prices and quantities toward market clearing: the point where everything that's produced by a seller is sold to a buyer. And this equilibrium idea is what gave us a price and quantity.

We've only ever looked at a single market. But in the background we know there must be markets for all the goods and services we're considering. So going back to the farmers, we can apply our notion of markets to tell us something about prices and quantities for spinach and carrots.

Remember we found an exchange rate for trading between the two farmers? We said the exchange rate allowed the farmers to not just co-op, but exchange with each other as separate entities while still benefitting from specialization and trade. We said this exchange rate must live between their own internal exchange rates, represented by their PPFs. We didn't decide which one was best, but said that there were a range of prices that would make the trade benefit both farmers.

What we're doing now with markets pins this exchange rate down. Using a market for spinach and a separate market for carrots, we have buyers and sellers making up the demand and supply curves, all facing the incentives that push the market toward equilibrium. The equilibrium generates prices in both markets. And these prices coordinate the decisions of the farmers by forming an exchange rate that allows the farmer to specialize accordingly.

This is some of what's happening in the background for the farmer, who makes up part of the supply curve. If the farmer is going to produce a crop (and we'll come back to exactly how the farmer would think about it), their decision about WHICH crop comes from their own production functions and the prices in the markets. When prices are relatively high in the carrot market, both farmers will grow carrots. And the same goes for spinach. But when the prices live somewhere

in the middle, between the two farmer's opportunity costs, then one will grow one crop and the other the other.

And since there are many farmers in the market with different PPFs (and hence different production functions), when working effectively, the market will arrive at a pair of prices that supply the appropriate quantities to the market once every farmer has decided their crop choices.

Closing

{++We've found where prices come from. Next time we keep the same setup and ask how good the market's answer is: we'll add up the gains from every exchange, and see what happens when a government moves the price.++}
